

SCANNING AND COLLECTION OF POLLUTED ITEMS IN WATER BODIES BY

TEAM:

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THEME: PURIFIES WATER, RESTORING LIFE" - EMPHASIZES THE IMPORTANCE OF CLEAN WATER FOR "REMOVING POLLUTION, REVIVING ECOSYSTEMS - HIGHLIGHTS THE IMPACT OF POLLUTION REMOVAL ON ECOSYSTEMS.

Problem Statement



REAL LIFE VALIDATION

- WHAT IS THE PROBLEM STATEMENT
- REMOVING THE POLLUTED ITEMS FROM WATER BODIES
- IT IS NOT GOOD TO THE ENVIRONMENT
- IT IS THE REASON FOR THE ENVIRONMENTAL DEGRADING.

CAUSES OF WATER POLLUTION



EXAMPLE:
 DEFORESTATION
 1.OIL POLLUTION
 2.AGRICULTURAL
 WASTE
 3.RADIOACTIVE WASTE
 4.WASTE WATER
 5.THERMAL
 POLLUTION4.

Solution



- **SOLUTION: IMPLEMENTING ADVANCED WATER FILTRATION SYSTEMS AND BIOREMEDIATION TECHNIQUES TO REMOVE POLLUTANTS AND RESTORE AQUATIC ECOSYSTEMS.**
- **KEY FEATURES: REAL-TIME MONITORING, SUSTAINABLE PRACTICES, AND COMMUNITY ENGAGEMENT TO ENSURE LONG-TERM WATER QUALITY IMPROVEMENT.**

USP/Differentiator

Feature Matrix: Scanning and Collection of Polluted Items on Water Bodies

Method	Technology Used	Features	Advantages	Limitations
Manual Cleanup	Human labor, basic tools (nets, boats)	Targeted collection, flexible, low-tech	Low cost, adaptable, good for community involvement	Labor-intensive, slow, limited coverage
Drone Surveillance	Aerial drones with cameras & sensors Would you like a visual version of this or a focused breakdown by technology or use-case (e.g., urban rivers)	Real-time imagery, GPS mapping, AI for detection	Large area coverage, fast scanning, remote access	Expensive, weather dependent, no collection ability
Autonomous Surface Vehicles (ASVs)	GPS, sonar, AI, sensors	Automated scanning and collection, data logging	Continuous operation, precise collection	High cost, complex maintenance
Booms and Barriers	Floating barriers	Passive collection, prevents spread	Simple, effective for floating trash	Limited to surface pollutants, needs maintenance
Smart Trash Skimmers	AI, solar-powered motors, sensors	Autonomous collection, trash detection, self-charging	Eco-friendly, scalable, low maintenance	High initial cost, limited to floating debris
Satellite Monitoring	Remote sensing, multispectral imagery	Large-scale scanning, pollution trend analysis	High-altitude coverage, historical data tracking	Low resolution for small debris, delayed data
Underwater Drones (ROVs)	Cameras, sonar, robotic arms	Detects submerged items, precise manipulation	Access to deeper pollutants, detailed inspection	High operational cost, limited range
Water Quality Sensors	Chemical, biological, and physical sensors	Detect pollutants (not physical debris), monitor toxicity	Real-time data, long-term monitoring	Doesn't remove trash, only detects pollutants



The impact of a project can be significant, bringing about various benefits and changes. Some potential impacts include:

1. Improved efficiency
2. Increased productivity
3. Enhanced customer satisfaction
4. Cost savings
5. New opportunities
6. Skill development
7. Competitive advantage

What specific impact are you looking for? The impact of a project can be significant, bringing about various benefits and changes. Some potential impacts include:

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Project Plan/Methodology

PLANNING:

- 1. Objective: Restore water quality and ecosystem balance.
- 2. Assessment: Identify pollutants and sources.
- 3. Solution: Implement filtration and bioremediation.
- 4. Timeline: Set milestones and deadlines.
- 5. Budget: Allocate resources and function

METHODOLOGY:

1. Assessment: Evaluate water quality and identify pollutants.
2. Solution Design: Develop tailored restoration plan.
3. Implementation: Install filtration systems and bioremediation techniques.
4. Monitoring: Track progress and adjust as needed.
5. Maintenance: Ensure long-term sustainability

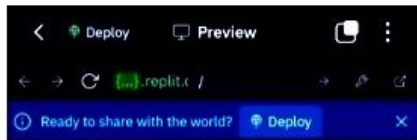


PROTOTYPE OF OUR PROJECT

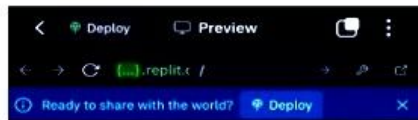


Location Description (optional)

Analyze for Pollution



Source: [Source]



Pollution Analysis Results

Total Pollutants Detected

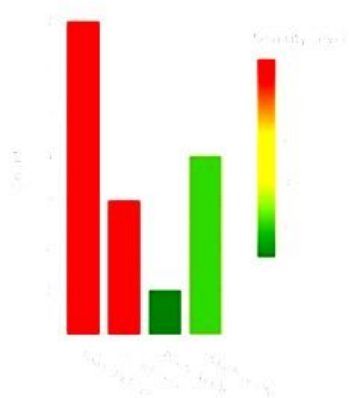
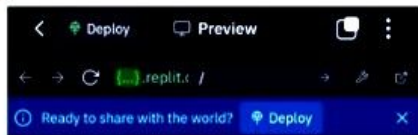
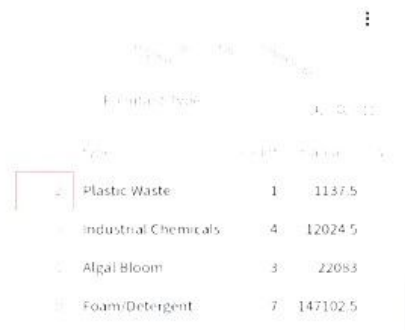
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Pollution Density

48.66 per M px

Pollutant Types

4

Waste Type	Count
Plastic Waste	1137.5
Industrial Chemicals	12024.5
Algal Bloom	22083
Foam/Detergent	147102.5

TEAM

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Conclusion



IMPLEMENTING INNOVATIVE WATER RESTORATION SOLUTIONS, WE CAN REVITALIZE POLLUTED WATER BODIES, SUPPORT AQUATIC LIFE, AND PROMOTE SUSTAINABLE ECOSYSTEMS FOR FUTURE GENERATIONS."

HOW IT IS DIFFERENT FROM OTHER SOLUTIONS:

1. Real-Time Environmental Sensing Your System: Uses sensors (e.g., turbidity, chemical, temperature) to detect and localize pollutants on the spot. Other Apps: Often rely on delayed government or third-party data uploads.
2. Autonomous Operation Your Prototype: Employs AI or programmed logic to autonomously scan and clean without human intervention. Other Apps: Typically need manual input or only guide humans to pollution hotspots.

